Lecture 21: Runge's Example; Spline Spaces; B-Splines

Math 6610

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Today's Goals

- The spline space $\mathcal{S}_{k,\Delta}$ and its dimension
- Truncated powers
- B-splines

Recap from Lecture 20

$f^{(n+1)}$ continuous on $[a,b]$; p_n interpolates f at distinct $x_0, \ldots, x_n$; $\omega(x) = \prod_{i=0}^{n}(x - x_i)$; $M = \max_{[a,b]} \left|f^{(n+1)}\right|$.

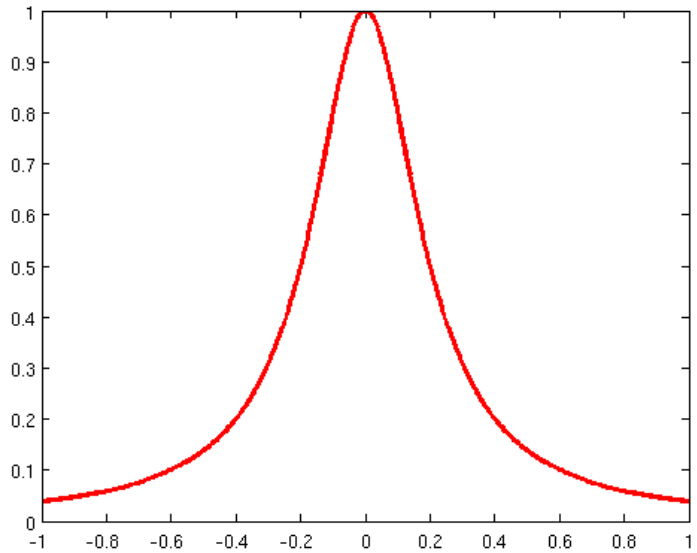
$$\max_{[a,b]} |f - p_n| \leqslant \underbrace{\frac{M}{(n+1)!}}_{\text{fixed by } f} \underbrace{\max_{[a,b]} |\omega|}_{\text{chosen by us}}$$

- Chebychev nodes make $\max_{[-1,1]} |\omega| = 2^{-n}$, the smallest possible. On $[a,b]$:

$$\max_{[a,b]} |f - p_n| \leqslant \frac{M}{(n+1)!} \cdot 2\left(\frac{b-a}{4}\right)^{n+1}.$$

Equally Spaced vs. Chebychev Nodes

High degree polynomials using equispaced points suffer from many oscillations



- Chebychev nodes: the error is distributed more evenly

Runge's Example: The Experiment

$f(x) = 1/(1+25x^2)$ on $[-1,1]$, degree n, both node sets:

```
f  = @(x) 1./(1 + 25*x.^2);
xx = linspace(-1, 1, 20001);                    % fine grid
for n = [5 10 20]
  xe = linspace(-1, 1, n+1);                    % equally spaced
  xc = cos(pi*(2*(0:n) + 1)/(2*n + 2));         % Chebychev
  pe = polyfit(xe, f(xe), n);  pc = polyfit(xc, f(xc), n);
  ee = max(abs(f(xx) - polyval(pe, xx)));
  ec = max(abs(f(xx) - polyval(pc, xx)));
  fprintf('%3d  %9.3g  %9.3g\n', n, ee, ec)
end
```

Remark. `polyfit` uses monomials ($\kappa_2 \approx 8.3 \cdot 10^8$ at $n = 20$, equally spaced; Lecture 17), but the errors on the next slide are still correct to the digits shown.

Runge's Example

Example. Maximum error $\max_{[-1,1]} |f - p_n|$ (on the fine grid):

n	5	10	20
equally spaced	0.43	1.92	59.8
Chebychev	0.56	0.11	0.015

- Equally spaced: the error *grows* with n (near ± 1).
- Chebychev: the error *decreases*.

Question: what does the error bound say about this f?

Runge's Example

1. M is huge. For $|x| < 1/5$, $f(x) = \sum_{m \geqslant 0} (-25x^2)^m$, so for even m

$$\frac{\left|f^{(m)}(0)\right|}{m!} = 25^{m/2} = 5^m \quad \Longrightarrow \quad \frac{M}{(n+1)!} \geqslant 5^{n+1} \quad (n \text{ odd}).$$

2. The Chebychev bound grows. With $b - a = 2$:

$$\frac{M}{(n+1)!} \cdot 2\left(\frac{2}{4}\right)^{n+1} \geqslant 5^{n+1} 2^{-n} \to \infty.$$

3. Yet the Chebychev error decreases. A bound is only an upper bound. (Fact, proof omitted: for Lipschitz continuous f, Chebychev interpolants converge uniformly on $[-1, 1]$.)

From Degree to Order

Given: $\Delta = \{x_0, \ldots, x_n\}$, $a = x_0 < x_1 < \cdots < x_n = b$, $n \geqslant 1$ (the knots t_i of Lectures 17–19).

Definition (*Spline of order k*). Let $k \geqslant 2$. S is a *spline of order k* on Δ if

(i) $S \in C^{k-2}[a, b]$, and

(ii) on each $[x_{j-1}, x_j]$, $j = 1, \ldots, n$, S is in $\prod_{k-1}$ (degree $\leqslant k-1$).

$\mathcal{S}_{k,\Delta}$:= the set of all such S.

order k	degree $k-1$	name (Lectures 17–19)
2	1	degree 1 spline, C^0
3	2	quadratic spline, C^1
4	3	cubic spline, C^2

Remark. $\mathcal{S}_{k,\Delta}$ is a real vector space (closed under sums and real multiples), and $\prod_{k-1} \subset \mathcal{S}_{k,\Delta}$.

The Dimension of $\mathcal{S}_{k,\Delta}$

Proposition. $\dim \mathcal{S}_{k,\Delta} = k + n - 1$.

Remark (*Why this number: a count*). First interval: k coefficients. Each of the $n-1$ next intervals: k new coefficients, $k-1$ matching conditions at its left knot. Total:

$$k + (n-1)\big(k - (k-1)\big) = k + n - 1.$$

Not a proof: it assumes the matching conditions are independent.

Remark (*Link to Lecture 18*). Interpolation fixes $n+1$ conditions; $k-2$ remain free. Cubic ($k = 4$): the 2 end conditions.

Proof plan: exhibit a basis with $k + n - 1$ elements (next three slides).

Truncated Powers

Definition (*Truncated power*). For a knot x_j and $k \geqslant 2$,

$$(x - x_j)_+^{k-1} := \begin{cases} (x - x_j)^{k-1}, & x \geqslant x_j, \\ 0, & x < x_j. \end{cases}$$

Lemma. $(x - x_j)_+^{k-1} \in C^{k-2}(\mathbb{R})$, and its $(k-1)$-st derivative jumps by $(k-1)!$ at x_j. Hence $(x - x_j)_+^{k-1} \in \mathcal{S}_{k,\Delta}$ for $j = 1, \ldots, n-1$.

Proof. 1. Away from x_j, for $l = 0, \ldots, k-2$:

$$\frac{d^l}{dx^l}(x - x_j)_+^{k-1} = (k-1)\cdots(k-l)\,(x - x_j)_+^{k-1-l}.$$

2. At x_j, induction on l: the difference quotient of $c\,(x - x_j)_+^{k-l}$ at x_j is $c\,\eta_+^{k-l}/\eta \to 0$ ($k - l \geqslant 2$). So the formula holds at x_j too, and is continuous.
3. Order $k-1$*:* 0 for $x < x_j$, $(k-1)!$ for $x > x_j$. ■

The Truncated Power Basis (1/2)

Theorem. The $k+n-1$ functions

$$B = \{1, x, \ldots, x^{k-1},\ (x-x_1)_+^{k-1}, \ldots, (x-x_{n-1})_+^{k-1}\}$$

form a basis of $\mathcal{S}_{k,\Delta}$.

Proof. $B \subset \mathcal{S}_{k,\Delta}$ by the lemma. Show: independent, then spanning.
1. Independence: setup. Suppose, for all $x \in [a,b]$,

$$S(x) := \sum_{l=0}^{k-1} \alpha_l x^l + \sum_{i=1}^{n-1} \beta_i (x-x_i)_+^{k-1} = 0.$$

2. Jump at x_j, $1 \leqslant j \leqslant n-1$. Near x_j, every term except the j-th is a polynomial, so only that term makes $S^{(k-1)}$ jump. By the lemma,

$$0 = S^{(k-1)}(x_j^+) - S^{(k-1)}(x_j^-) = \beta_j (k-1)! \quad \Longrightarrow \quad \beta_j = 0.$$

(continued on the next slide)

The Truncated Power Basis (2/2)

Proof (continued). 3. Monomials. Now $\sum_l \alpha_l x^l = 0$ on $[a, b]$: infinitely many zeros, so all $\alpha_l = 0$. Hence B is independent.
4. Spanning: subtract the jumps. Let $S \in \mathcal{S}_{k,\Delta}$, $p \in \prod_{k-1}$ with $p = S$ on $[x_0, x_1]$, and c_j the jump of $S^{(k-1)}$ at x_j. Set

$$R := S - p - \sum_{j=1}^{n-1} \frac{c_j}{(k-1)!}(x - x_j)_+^{k-1} \in \mathcal{S}_{k,\Delta}.$$

5. $R = 0$, induction over the intervals.

- Base: $R = 0$ on $[x_0, x_1]$; and $R^{(k-1)}$ has no jumps.
- Step: $R = 0$ on $[x_{j-1}, x_j] \Rightarrow R, \ldots, R^{(k-1)}$ vanish at x_j from both sides $\Rightarrow$ (Taylor, $R \in \prod_{k-1}$) $R = 0$ on $[x_j, x_{j+1}]$.

So $S = p + \sum_j \frac{c_j}{(k-1)!}(x - x_j)_+^{k-1} \in \operatorname{span} B$. ■

Remark. Hence $\dim \mathcal{S}_{k,\Delta} = k + n - 1$.

Truncated Powers in Practice

Not-a-knot cubic (Lecture 18), $n \geqslant 3$: knots $x_2, \ldots, x_{n-2}$, so

$$1,\ x,\ x^2,\ x^3,\ (x - x_j)_+^3,\ j = 2, \ldots, n-2 \qquad (n+1 \text{ functions}, n+1 \text{ data}).$$

Interpolation matrix T (row r: the basis at x_{r-1}):

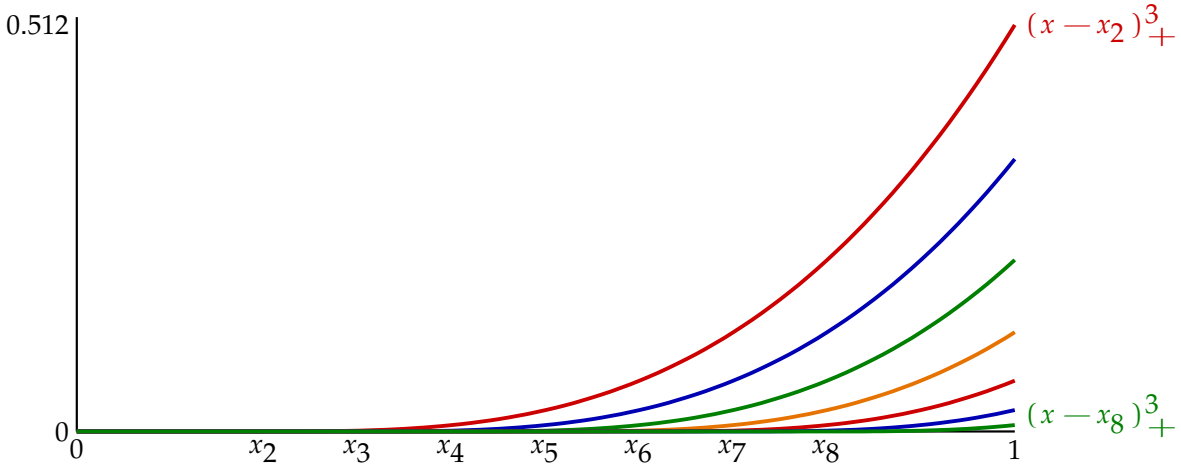
```
n = 20;   x = linspace(0, 1, n+1)';              % equally spaced
T = [x.^0   x   x.^2   x.^3   max(x - x(3:n-1)', 0).^3];
cond(T, inf)
```

n	5	10	20	40	80
$\kappa_\infty(T)$	$2.9 \cdot 10^3$	$4.2 \cdot 10^4$	$4.9 \cdot 10^5$	$6.4 \cdot 10^6$	$9.2 \cdot 10^7$

At $n = 80$: up to ≈ 8 digits lost (Lecture 15: relative error up to $\approx \kappa \cdot \epsilon_{\text{mach}}$).

Truncated Powers: A Picture

$(x - x_j)_+^3$, $x_j = j/10$, $j = 2, \ldots, 8$, on $[0, 1]$:



- Each curve is nonzero from x_j all the way to $b = 1$: global support.
- Neighbours look alike (same shape, shifted by 0.1), and the right ones are tiny ($(x - x_8)_+^3 \leqslant 0.008$): nearly dependent columns of very different size.

What Goes Wrong, and What We Want

Two defects of truncated powers:

- *Ill-conditioned:* for nearby knots, $(x - x_j)_+^3$ and $(x - x_{j+1})_+^3$ nearly coincide $\Rightarrow$ columns of T nearly parallel.
- *Global support:* $(x - x_j)_+^3 \neq 0$ on all of $(x_j, b] \Rightarrow T$ dense below the diagonal; $S(x)$ needs $O(n)$ terms.

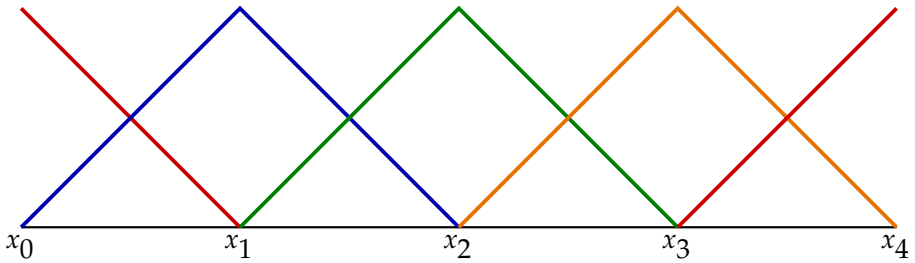
Wish list for a basis of $\mathcal{S}_{k,\Delta}$:

- *local:* each function nonzero on at most k consecutive intervals;
- *nonnegative*, and *summing to* 1 at every $x \in [a, b]$;
- *well conditioned*, cheap and stable to evaluate.

Answer: B-splines. Defined today; the wish list is proved and measured in Lecture 22.

The Simplest Case: Hat Functions ($k = 2$)

On $x_0, \ldots, x_4$: N_{i2} is 1 at x_{i-1}, 0 at the other knots, linear in between.



- Local: each hat lives on $\leqslant 2$ intervals.
- Nonnegative, and the hats sum to 1.
- Interpolation needs no linear system:

$$S(x) = \sum_{i=1}^{n+1} y_{i-1} N_{i2}(x) \quad \Longrightarrow \quad S(x_{i-1}) = y_{i-1}, \ i = 1, \ldots, n+1.$$

Goal: the same for any order k.

Knots for B-Splines

A *knot sequence* may repeat values:

$$\tau_1 \leqslant \tau_2 \leqslant \cdots \leqslant \tau_l, \qquad \tau_i \in \mathbb{R}. \tag{1}$$

Definition (*Multiplicity*). The *multiplicity* of a knot value is the number of times it appears in (1).

Definition (*Clamped knots for $\mathcal{S}_{k,\Delta}$*). End points k times, interior knots once:

$$\overbrace{\underbrace{x_0}_{\tau_1=\cdots=\tau_k}}^{k \text{ times}} < \underbrace{x_1}_{\tau_{k+1}} < \cdots < \underbrace{x_{n-1}}_{\tau_{k+n-1}} < \overbrace{\underbrace{x_n}_{\tau_{k+n}=\cdots=\tau_{2k+n-1}}}^{k \text{ times}}$$

Count: $l = 2k + n - 1$, and $l - k = k + n - 1 = \dim \mathcal{S}_{k,\Delta}$.

Example. $k = 2$, $n = 4$: $\tau = (x_0, x_0, x_1, x_2, x_3, x_4, x_4)$, $l = 7$: 5 hats (the two doubled end points give the two half hats).

B-Splines: Definition

Definition (*B-splines; Cox–de Boor recursion*). For the knots (1), the B-splines N_{ij} (index i, order j):

$$N_{i1}(x) := \begin{cases} 1, & \tau_i \leqslant x < \tau_{i+1}, \\ 0, & \text{otherwise}, \end{cases} \qquad i = 1, \ldots, l-1,$$

and for $j = 2, \ldots, k,\ \ i = 1, \ldots, l-j$:

$$N_{ij}(x) := \frac{x - \tau_i}{\tau_{i+j-1} - \tau_i} N_{i,j-1}(x) + \frac{\tau_{i+j} - x}{\tau_{i+j} - \tau_{i+1}} N_{i+1,j-1}(x). \tag{2}$$

In words: order j = (rising ramp) $\times$ left neighbour $+$ (falling ramp) $\times$ right neighbour, both of order $j-1$.

B-Splines: Two Conventions

1. A term with denominator 0 is 0. (Nothing is lost: $\tau_i = \tau_{i+j-1}$ forces $N_{i,j-1} \equiv 0$.)
2. $N_{ik}(\tau_l) := \lim_{x \to \tau_l^-} N_{ik}(x)$, so the right end point $b = \tau_l$ is covered.

Example (*Convention 1 at work: a half hat*). $k = 2$, clamped: $\tau_1 = \tau_2 = x_0 < \tau_3 = x_1$. In (2) with $i = 1$, $j = 2$:

$$N_{12}(x) = \underbrace{\frac{x - \tau_1}{\tau_2 - \tau_1} N_{11}(x)}_{\text{denominator } 0\text{: term } 0} + \frac{\tau_3 - x}{\tau_3 - \tau_2} N_{21}(x) = \begin{cases} \dfrac{x_1 - x}{x_1 - x_0}, & x_0 \leqslant x < x_1, \\ 0, & \text{otherwise.} \end{cases}$$

This is the left half hat: 1 at x_0, 0 at x_1. (And $N_{11} \equiv 0$ anyway: its box $[\tau_1, \tau_2)$ is empty.)

Orders 1 and 2 by Hand

Equally spaced knots $\tau_i = i$ (no repeats).
Order 1: boxes, $N_{i1} = 1$ on $[i, i+1)$, 0 otherwise.
Order 2: (2) with $j = 2$; both denominators are 1:

$$N_{i2}(x) = (x-i)\,N_{i1}(x) + (i+2-x)\,N_{i+1,1}(x) = \begin{cases} x-i, & i \leqslant x < i+1, \\ i+2-x, & i+1 \leqslant x < i+2, \\ 0, & \text{otherwise.} \end{cases}$$

Up-ramp on the left box $+$ down-ramp on the right box $=$ a hat.

Remark. The weight $\frac{x-\tau_i}{\tau_{i+j-1}-\tau_i}$ rises from 0 to 1 on $[\tau_i, \tau_{i+j-1}]$; the other falls from 1 to 0 on $[\tau_{i+1}, \tau_{i+j}]$. So each order adds one interval to the support (the set where the function is nonzero).

Order 3 by Hand

Same knots, $j = 3$; both denominators are 2. With $s = x - i$:

$$N_{i3}(x) = \tfrac{s}{2} N_{i2}(x) + \tfrac{3-s}{2} N_{i+1,2}(x) = \begin{cases} \frac{1}{2}s^2, & 0 \leqslant s < 1, \\ \frac{1}{2}(-2s^2 + 6s - 3), & 1 \leqslant s < 2, \\ \frac{1}{2}(3-s)^2, & 2 \leqslant s < 3, \\ 0, & \text{otherwise.} \end{cases}$$

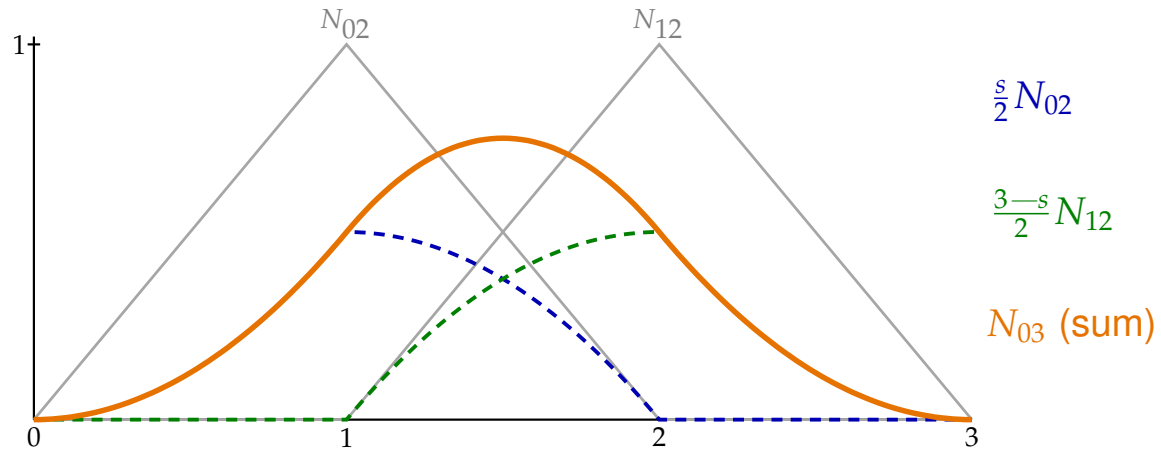
(Middle piece: $\frac{s}{2}(2-s) + \frac{3-s}{2}(s-1)$.)

Check the wish list:

- Smooth: C^1 at $s = 1$ (values $\frac{1}{2} = \frac{1}{2}$, slopes $1 = 1$); at $s = 2$ by symmetry $s \mapsto 3 - s$.
- Local: support $[i, i+3]$, 3 intervals.
- Sum 1: at $x = 1.5$ only $N_{-1,3}, N_{0,3}, N_{1,3}$ are nonzero ($s = 2.5, 1.5, 0.5$): $\frac{1}{8} + \frac{3}{4} + \frac{1}{8} = 1$.

Order 3: The Recursion as a Picture

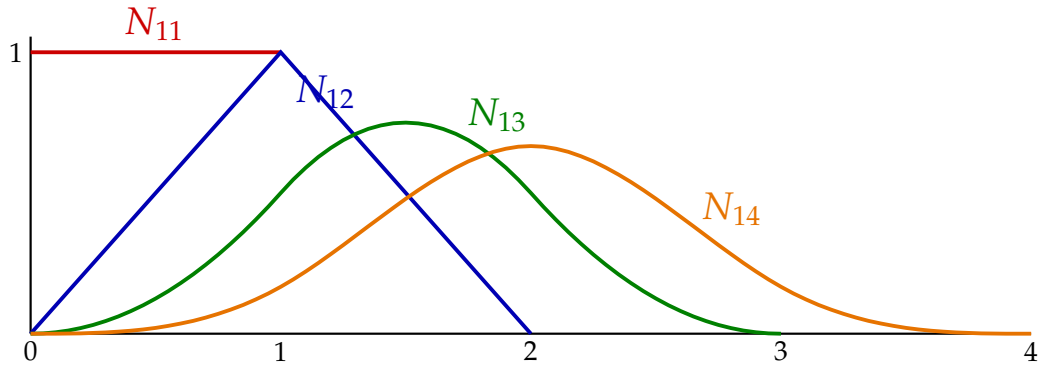
$i = 0$, $\tau_i = i$: $N_{03} = \frac{s}{2} N_{02} + \frac{3-s}{2} N_{12}$, $s = x$.



- Gray: the two hats of order 2. Dashed: each hat times its ramp.
- The sum (orange) is smoother than either term: the corners of the hats cancel, leaving C^1.
- Peak value $N_{03}(1.5) = \frac{3}{4}$. (Check: $\frac{1.5}{2} \cdot 0.5 + \frac{1.5}{2} \cdot 0.5 = 0.75$.)

Orders 1 to 4 on Equally Spaced Knots

$N_{11}, N_{12}, N_{13}, N_{14}$ for $\tau_i = i$: supports $[0,1]$, $[0,2]$, $[0,3]$, $[0,4]$.



- Each order: one more interval of support, one more continuous derivative.
- N_{14}, the cubic B-spline: C^2, with $N_{14} = N'_{14} = N''_{14} = 0$ at both ends of its support.

Looking Ahead

- **Next lecture:** B-splines: properties (the wish list, proved), fast evaluation, interpolation